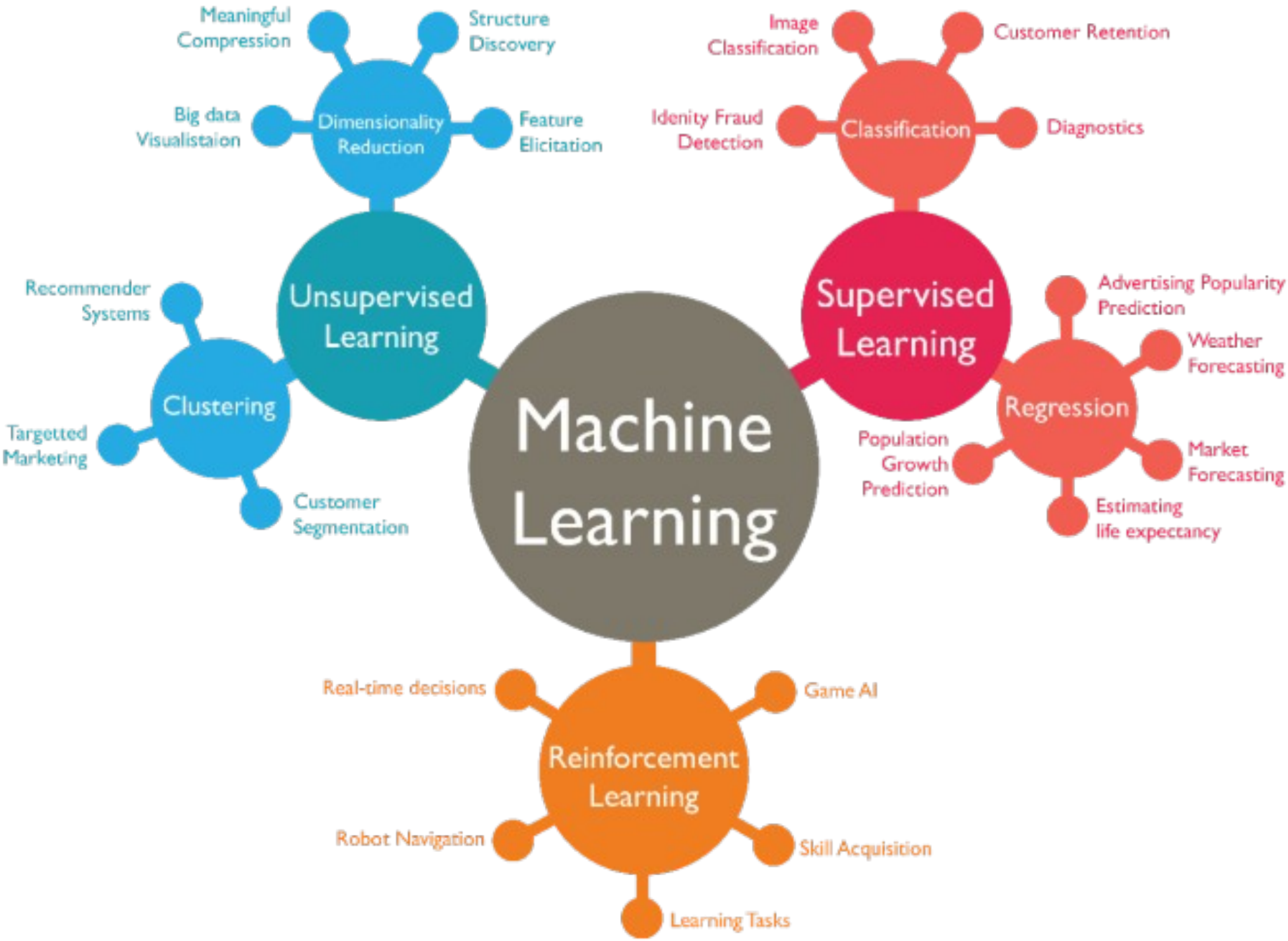
How to train the model?

What are the types of machine learning algorithms?


Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Reinforcement Learning
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

Unsupervised learning

This is complicated…

What does it mean?

We have many, many different entities

	x	y
0	8.622185392821123	1.9357957863943338
1	-4.736709578070003	-7.970957652334162
2	9.621222051741709	0.9254231466750656
3	6.1620950286859735	-0.27325436806911185
4	8.697488088714522	-1.0574520600618431
5	-4.635727957237028	-11.353826370491337
6	6.6165131565988915	1.148031347237489
7	-7.027641625085397	-10.134639291120877
8	6.159924603264129	1.4770241897988083
9	8.764907668651523	0.1443565342759367
10	-5.1579307061874955	-9.991878624558826
11	7.213336893425731	1.5105456994536408
12	8.255057886498266	0.13445014653935666
13	7.379562483316606	0.4011770735289152
14	5.942055856146272	10.507683327709145
15	7.883571427924204	10.939695145994671
16	7.829448164576652	9.626271582582962
17	-5.2662850574087905	-9.900710566089337
18	7.144610537616683	11.254631886978462
19	6.346814725561926	10.330756423082127
20	4.970850432267156	1.6942169921979364

We have a few entities, but characterized by many features

Country	Spirits	Wine	Beer	Life Exp.	Heart Dis.
France	2.5	63.5	40.1	78	61.1
Italy	0.9	58.0	25.1	78	94.1
Swiss	1.7	46.0	65.0	78	106.4
Austria	1.2	15.7	102.1	78	173.0
UK	1.5	12.2	100.0	77	199.7
US	2.0	8.9	87.8	76	176.0
Russia	3.8	2.7	17.1	69	373.6
Czech	1.0	1.7	140.0	73	283.7
Japan	2.1	1.0	55.0	79	34.7
Mexico	0.8	0.2	50.4	73	36.4

Our goal: to simplify

We have many, many different entities

-0.4909874269576609	0.92174625551858	cluster_1
-0.011950322950551207	-0.5028213130035346	cluster_2
0.017666875862201087	-0.9704153907324322	cluster_2
0.2894010092992185	-0.31838689049453583	cluster_2
0.5561826504568832	-0.11303322776415026	cluster_2
0.4358162536213022	0.1334397813626072	cluster_1
0.0979859734527061	-0.4790873405106578	cluster_2
0.4798033229751856	-0.7926836074206647	cluster_2
0.5962017592811004	0.8546833307470932	cluster_1
0.8835751288634599	-0.4403258352535013	cluster_2
0.17144166900967786	0.47079237590172063	cluster_1
0.45538153365676665	-0.0779659615036343	cluster_2
0.43448708140854814	-0.2591104592159882	cluster_2
0.3874300674401985	-0.3078079049107563	cluster_2
-0.5073225093386902	0.9245068367486952	cluster_1
0.9205870518663738	-0.44761121878441	cluster_2
0.44596694201693304	-0.21363871068380375	cluster_2
-0.06391258238990716	0.9266669913068191	cluster_1
-0.05885812977804834	-1.0476320809657675	cluster_2
0.40282386505651646	0.3192971457768794	cluster_1
0.4956304654410191	-0.05541205666513304	cluster_2

We have a few entities, but characterized by many features

Country	Spirits	Wine	Beer	Life Exp.	Heart Dis.
France	2.5	63.5	40.1	78	61.1
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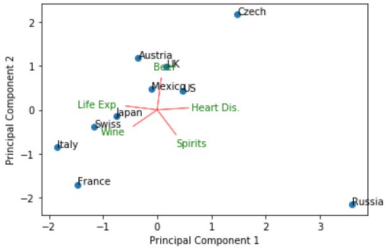
Czech	1.0	1.7	140.0	73	283.7
Japan	2.1	1.0	55.0	79	34.7
Mexico	0.8	0.2	50.4	73	36.4



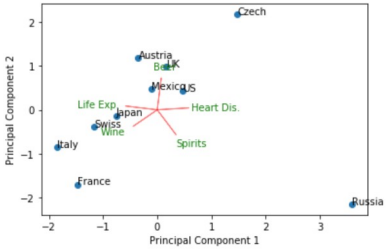
Why do we simplify?

Less differentiation

- Simpler to manage
- …

Lower numer of features

- Easier to understand
- …



Lower number of features

Dimensionality reduction

Alcohol consumption

Liters/year
Heart diseases: per 100 000 citizens/year

Country	Spirits	Wine	Beer	Life Exp.	Heart Dis.
France	2.5	63.5	40.1	78	61.1
Italy	0.9	58.0	25.1	78	94.1
Swiss	1.7	46.0	65.0	78	106.4
Austria	1.2	15.7	102.1	78	173.0
UK	1.5	12.2	100.0	77	199.7
US	2.0	8.9	87.8	76	176.0
Russia	3.8	2.7	17.1	69	373.6
Czech	1.0	1.7	140.0	73	283.7
Japan	2.1	1.0	55.0	79	34.7
Mexico	0.8	0.2	50.4	73	36.4

Source: Time Magazine, 1996 (Jan)

Trick: let's change our point of view...

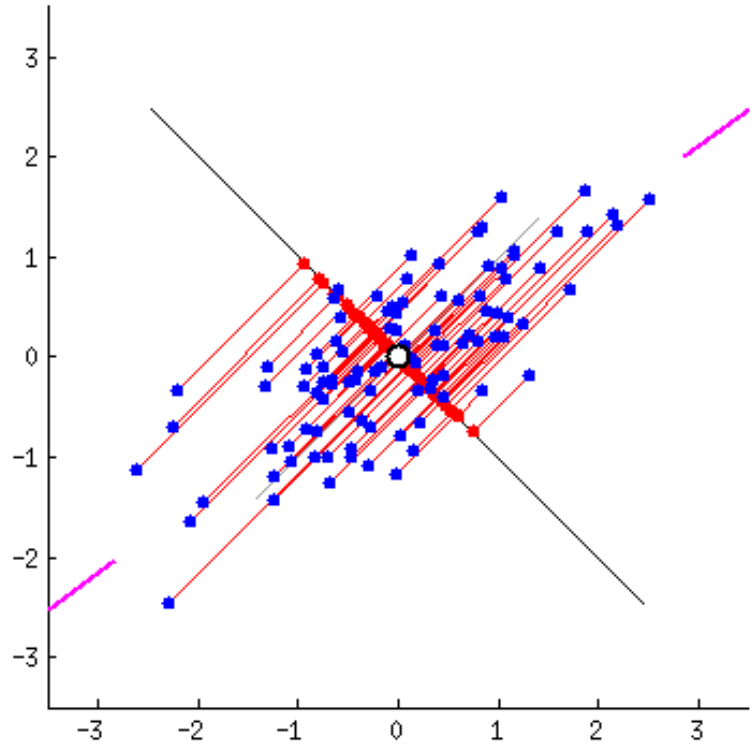
Let's just rotate the axes a little bit...

Before: each point is described by two variables

After:

- **Each point can be (roughly) described with one variable**
- **This new variable (feature) is a combination of original features ("rotation")**
- **Of course, we lose some information, but...**

This can be generalized to a lot of dimensions...



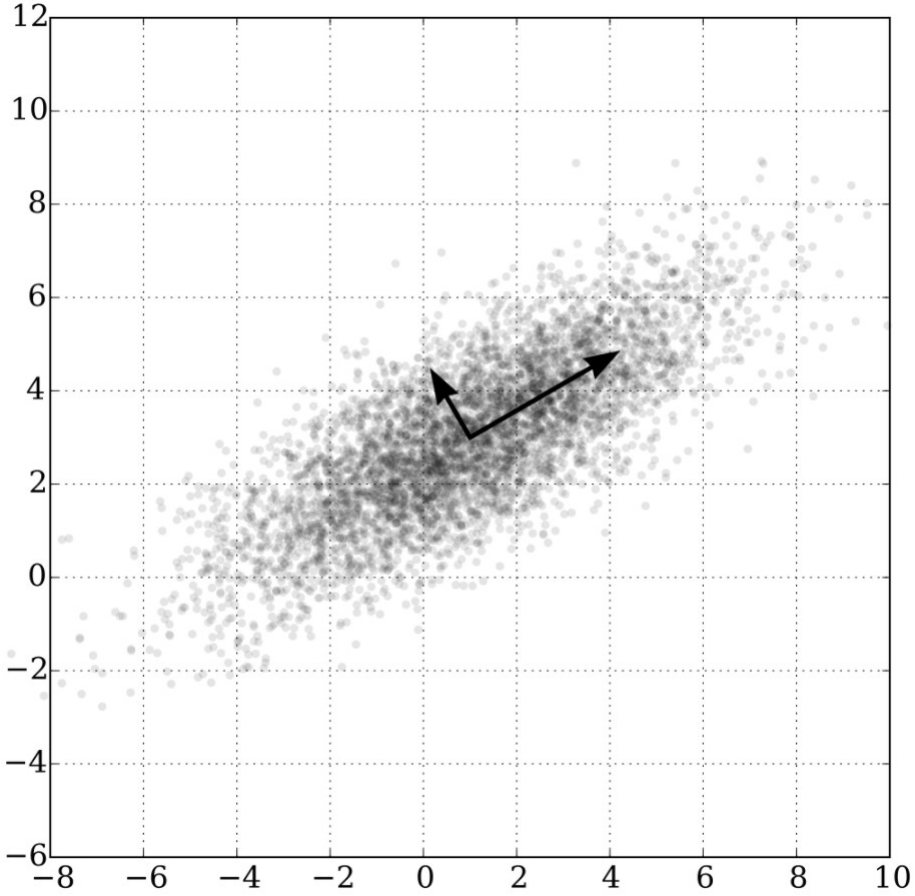
Source: https://stats.stackexchange.com/questions/2691/making-sense-of-principal-component-analysis-eigenvectors-eigenvalues

Dimensionality reduction

PCA – Principal Component Analysis

What for?

- **Easier to draw (less dimensions)**
- **Easier to understand**
- **Faster calculations (fewer features)**
- **... but also new findings.**

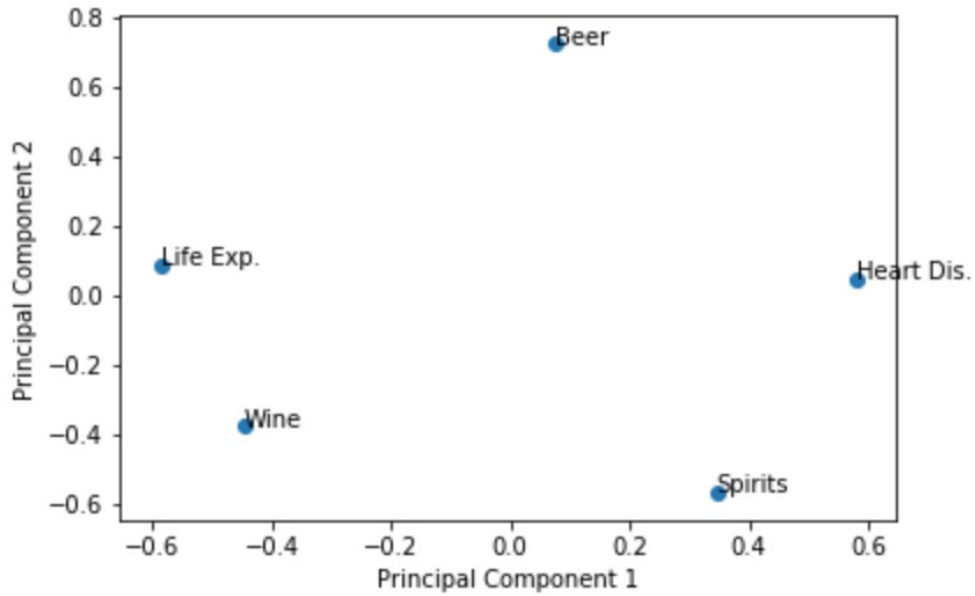


PCA: alcohol consumption

```
pca.explained_variance_ratio_
```

```
array([0.46027599, 0.32114766])
```

	Spirits	Wine	Beer	Life Exp.	Heart Dis.
PCA1	0.345896	-0.445039	0.073961	-0.584980	0.578467
PCA2	-0.568092	-0.378363	0.724405	0.086423	0.043377



PCA: alcohol consumption

Country	Spirits	Wine	Beer	Life Exp.	Heart Dis.
France	2.5	63.5	40.1	78	61.1
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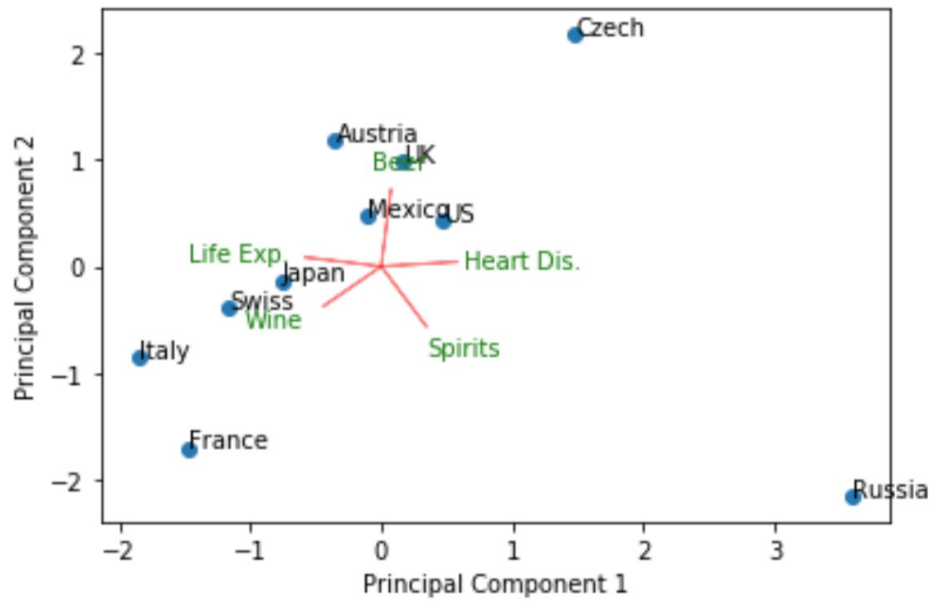
US	2.0	8.9	87.8	76	176.0
Russia	3.8	2.7	17.1	69	373.6
Czech	1.0	1.7	140.0	73	283.7
Japan	2.1	1.0	55.0	79	34.7
Mexico	0.8	0.2	50.4	73	36.4



This can work for millions of records (observations)...

country	beer_servings	spirit_servings	wine_servings	total_litres_of_pure_alcohol
afghanistan	0	0	0	0.0
albania	89	132	54	4.9
algeria	25	0	14	0.7
andorra	245	138	312	12.4
angola	217	57	45	5.9
antigua & barbuda	102	128	45	4.9
argentina	193	25	221	8.3
armenia	21	179	11	3.8
australia	261	72	212	10.4
austria	279	75	191	9.7
azerbaijan	21	46	5	1.3
bahamas	122	176	51	6.3
bahrain	42	63	7	2.0
bangladesh	0	0	0	0.0
barbados	143	173	36	6.3
belarus	142	373	42	14.4
belgium	295	84	212	10.5
belize	263	114	8	6.8
benin	34	4	13	1.1
bhutan	23	0	0	0.4
bolivia	167	41	8	3.8
bosnia-herzegovina	76	173	8	4.6
botswana	173	35	35	5.4
brazil	245	145	16	7.2
brunei	31	2	1	0.6
bulgaria	231	252	94	10.3
burkina faso	25	7	7	4.3
burundi	88	0	0	6.3
cote d'ivoire	37	1	7	4.0
cabo verde	144	56	16	4.0
cambodia	57	65	1	2.2
cameroon	147	1	4	5.8
canada	240	122	100	8.2
central african republic	17	2	1	1.8
chad	15	1	1	0.4
chile	130	124	172	7.6
china	79	192	8	5.0
colombia	159	76	3	4.2
comoros	1	3	1	0.1
congo	76	1	9	1.7
cook islands	0	254	74	5.9
costa rica	149	87	11	4.4
croatia	230	87	254	10.2
cuba	93	137	5	4.2
cyprus	192	154	113	8.2
czech republic	361	170	134	11.8
north korea	0	0	0	0.0
dr congo	32	3	1	2.3
denmark	224	81	278	10.4
djibouti	15	44	3	1.1
dominica	52	286	26	6.6
dominican republic	193	147	9	6.2
ecuador	162	74	3	4.2
egypt	6	4	1	0.2
el salvador	52	69	2	2.2

... and can also work for you

Features?

Customers? Products?

country	beer_servings	spirit_servings	wine_servings	total_litres_of_pure_alcohol	beer_servings	spirit_servings	wine_servings	total_litres_of_pure_alcohol
afghanistan	0	0	0	0.0	0	0	0	0.0
albania	89	132	54	4.9	89	132	54	4.9
algeria	25	0	14	0.7	25	0	14	0.7
andorra	245	138	312	12.4	245	138	312	12.4
angola	217	57	45	5.9	217	57	45	5.9
antigua & barbuda	102	128	45	4.9	102	128	45	4.9
argentina	193	25	221	8.3	193	25	221	8.3
armenia	21	179	11	3.8	21	179	11	3.8
australia	261	72	212	10.4	261	72	212	10.4
austria	279	75	191	9.7	279	75	191	9.7
azerbaijan	21	46	5	1.3	21	46	5	1.3
bahamas	122	176	51	6.3	122	176	51	6.3
bahrain	42	63	7	2.0	42	63	7	2.0
bangladesh	0	0	0	0.0	0	0	0	0.0
barbados	143	173	36	6.3	143	173	36	6.3
belarus	142	373	42	14.4	142	373	42	14.4
belgium	295	84	212	10.5	295	84	212	10.5
belize	263	114	8	6.8	263	114	8	6.8
benin	34	4	13	1.1	34	4	13	1.1
bhutan	23	0	0	0.4	23	0	0	0.4
bolivia	167	41	8	3.8	167	41	8	3.8
bosnia-herzegovina	76	173	8	4.6	76	173	8	4.6
botswana	173	35	35	5.4	173	35	35	5.4
brazil	245	145	16	7.2	245	145	16	7.2
brunei	31	2	1	0.6	31	2	1	0.6
bulgaria	231	252	94	10.3	231	252	94	10.3
burkina faso	25	7	7	4.3	25	7	7	4.3
burundi	88	0	0	6.3	88	0	0	6.3
cote d'ivoire	37	1	7	4.0	37	1	7	4.0
cabo verde	144	56	16	4.0	144	56	16	4.0
cambodia	57	65	1	2.2	57	65	1	2.2
cameroon	147	1	4	5.8	147	1	4	5.8
canada	240	122	100	8.2	240	122	100	8.2
central african republic	17	2	1	1.8	17	2	1	1.8
chad	15	1	1	0.4	15	1	1	0.4
chile	130	124	172	7.6	130	124	172	7.6
china	79	192	8	5.0	79	192	8	5.0
colombia	159	76	3	4.2	159	76	3	4.2
comoros	1	3	1	0.1	1	3	1	0.1
congo	76	1	9	1.7	76	1	9	1.7
cook islands	0	254	74	5.9	0	254	74	5.9
costa rica	149	87	11	4.4	149	87	11	4.4
croatia	230	87	254	10.2	230	87	254	10.2
cuba	93	137	5	4.2	93	137	5	4.2
cyprus	192	154	113	8.2	192	154	113	8.2
czech republic	361	170	134	11.8	361	170	134	11.8
north korea	0	0	0	0.0	0	0	0	0.0
dr congo	32	3	1	2.3	32	3	1	2.3
denmark	224	81	278	10.4	224	81	278	10.4
djibouti	15	44	3	1.1	15	44	3	1.1
dominica	52	286	26	6.6	52	286	26	6.6
dominican republic	193	147	9	6.2	193	147	9	6.2
ecuador	162	74	3	4.2	162	74	3	4.2
egypt	6	4	1	0.2	6	4	1	0.2
el salvador	52	69	2	2.2	52	69	2	2.2

Source: https://www.ty-penguin.org.uk/~auj/old_aber_pages/talks/depttalk96/pca.active.html

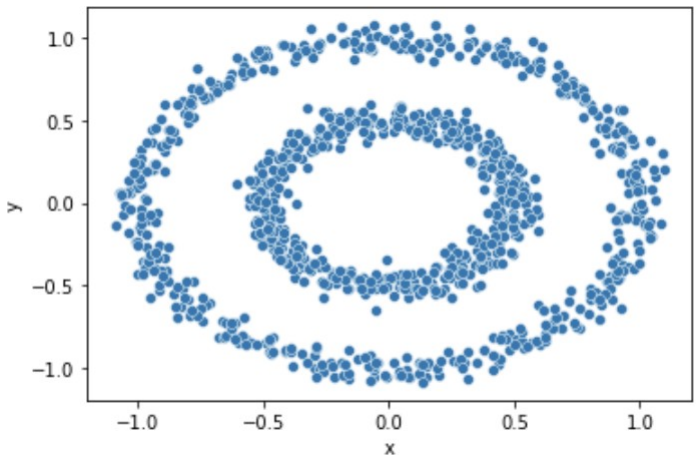
Less differentiation

Segmentation (clustering)

Look... What can you see?

1 000 objects

Each has 2 (numerical) features: x i y

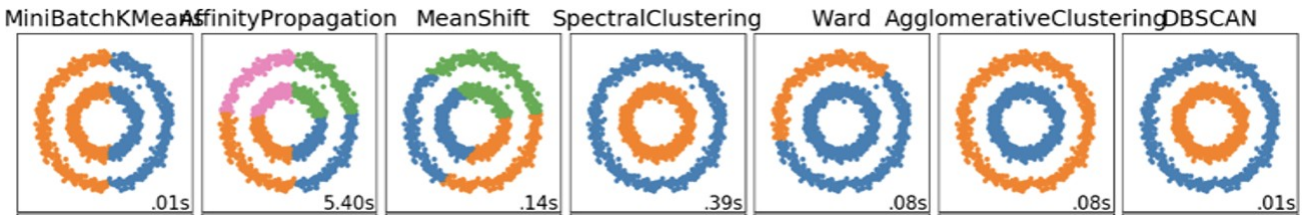


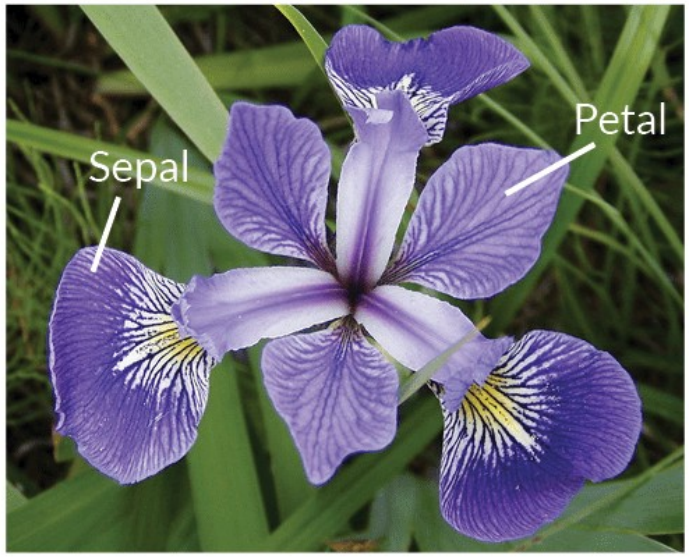
Look... What can you see?

1 000 objects
Each has 2 (numerical) features: x i y

Different ways of seeing things

Many algorithms, many patterns





Iris Versicolor

Iris Setosa

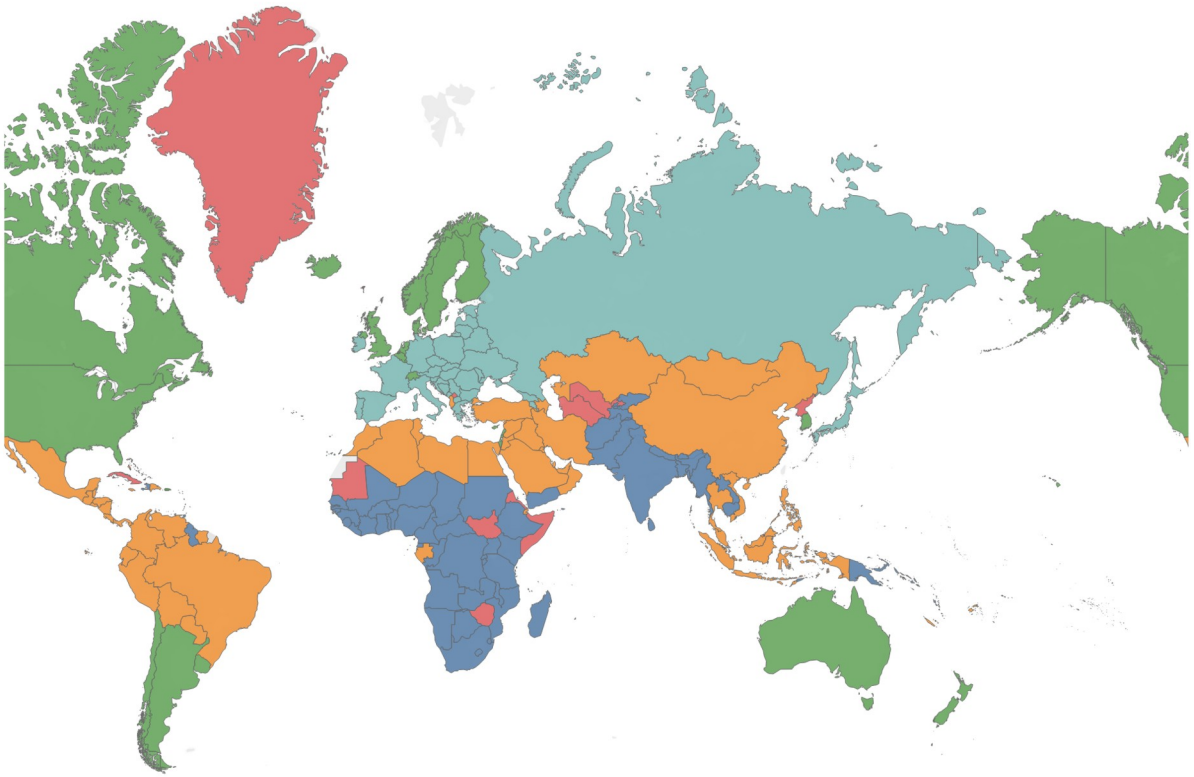
Iris Virginica

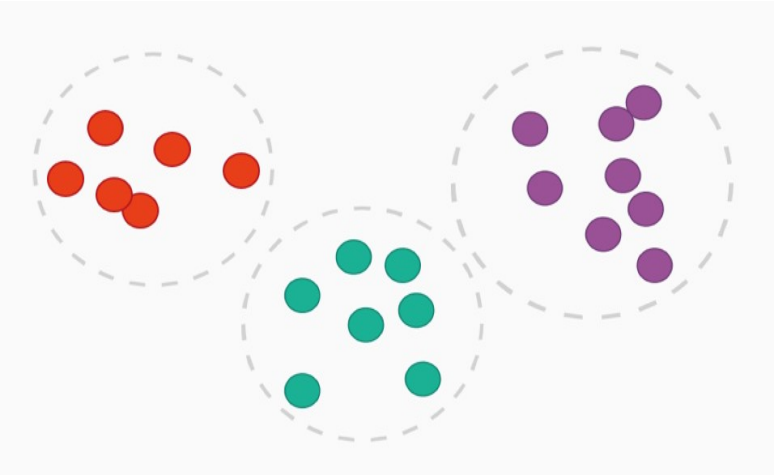
Example: flower segmentation

ID	sepal.length	sepal.width	petal.length	petal.width
string Integer	string Decimal	string Decimal	string Decimal	string Decimal
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
6	4.6	3.4	1.4	0.3
7	5.0	3.4	1.5	0.2
8	4.4	2.9	1.4	0.2
9	4.9	3.1	1.5	0.1
10	5.4	3.7	1.5	0.2
11	4.8	3.4	1.6	0.2
12	4.8	3.0	1.4	0.1
13	4.3	3.0	1.1	0.1
14	5.8	4.0	1.2	0.2
16	5.4	3.9	1.3	0.4
17	5.1	3.5	1.4	0.3
18	5.7	3.8	1.7	0.3
19	5.1	3.8	1.5	0.3
20	5.4	3.4	1.7	0.2
21	5.1	3.7	1.5	0.4

Example

Where should we promote our touristic offerings?



Unsupervised learning

Without a teaching set (set of known questions and answers)